

Take Home Quiz #1

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① ②

For Marci $I_0^{\text{Marci}} = I_0^{\text{Mickey}}$
 I_0 initial amount invested (in \$) @ 1/1/2018

$$I_{2018}^{2019} = I_0 + I_0(0.20) = I_1$$

$$I_{2019}^{2020} = I_1 - I_1(0.17) = I_2 = (1.20)I_0 - \text{gained } (0.20)I_0$$

$$= (I_0 + I_0(0.20)) - (0.17)(I_0 + I_0(0.20))$$

$$= I_0(1.20) - I_0(0.17 + 0.034)$$

$$= I_0(1.20 - 0.204)$$

$$I_2 = (0.996)I_0 : \text{lost } (0.004)I_0$$

* Marci's income from 2018-2019 was $(0.20)I_0$
but her income from 2019-2020 was $(-0.004)I_0$,

which ends up being
a loss for Marci,
that year.

* Thus, Marci earned more money in FY 2018-2019
than in FY 2019-2020. QED

② For Mickey

$$I_{2018}^{2019} = I_0 - (0.15)I_0 = I_1 = (0.85)I_0 : \text{lost } (0.15)I_0$$

$$I_{2019}^{2020} = I_1 + (0.25)I_1 = I_2$$

$$= (I_0 - (0.15)I_0) + (0.25)[I_0 - (0.15)I_0] = (0.85)I_0 + (0.25)I_0 - (0.0375)I_0 = (1.0625)I_0 : \text{gained } (0.0625)I_0$$

∴ Mickey earned
more money in FY 2019-2020
than in FY 2018-2019. QED

④ By 1/1/2020, Mickey's ^{net} portfolio investment was worth MORE THAN Marci's ^{net} portfolio investment.

Mickey's ^{Final} Portfolio Value	Marci's ^{Final} Portfolio Value
$(1.0625)I_0$	$(0.996)I_0$

⑤ Mickey's Final Portfolio Value was worth 6.65% MORE than Marci's Final Portfolio Value by 1/1/2020. QED

② $C_t = a + (b * t)$

① I would put " t " in terms of the end of the " n " day which the COVID-19 cases were counted.

This also allows for a C_0 base value, which can be transposed to other selected days of the COVID-19 data set.

② " b " would represent the growth rate of new COVID cases measured per day.

③ " a " would represent the # of ^{new} COVID cases measured BEFORE the END of the (1st) day in a selected data set. not necessarily zero.

④ Find " b " for 4/15/2020 Through 4/16/2020:

$$b = \frac{985}{(E85 - G85)}$$

(in terms of Excel ledger cells)

$$b = \frac{29,698.857}{(652,591 - 29,698.857)} = 0.0477$$

⑤ $a = 622,892.143$; $b = 0.0477 \rightarrow C_t = 622,892.143 + (0.0477)t$

QED

c) $a = E329 - G329$ <in Excel ledger cell terms>

$$a = 16,810,019 - 213,877.571$$

$$a \approx 16,596,141.429$$

$$b = G329$$

$$(E329 - G329)$$

$$= 213,877.571$$

$$16,596,141.429$$

$$\approx 0.0129$$

$$\textcircled{c} C_t^{12/15/20} = 16,596,141.429 + (0.0129)t$$

d) No! it assumes a linear growth rate for a PANDEMIC DISEASE where pandemics/epidemics usually have an exponential growth rate of infection.

* Eq ①'s linear approximation of $C_{245}^{12/15/20}$?

$$C_{245} = 622,892.143 + (0.047t)(245)$$

$$\approx 622,903.830 \text{ cases reported (projected)}$$

* Actual $C_{245}^{12/15/20} = 16,810,019$ cases reported (actual)

* Thus, the linear growth rate model from Eq. 1 was inaccurate by

$$\left(\frac{\text{cases projected} - \text{cases actual}}{\text{cases actual}} \right) \star 100 = \left(\frac{622,903.830 - 16,810,019}{16,810,019} \right) \star 100$$

Note! For the same data set this exp growth formula is MUCH more accurate

$$A_{02}^{12} = 622,892.143e^{(0.0133)t}$$

Percent Error of Eq. 1

$\star 100 = -96.294\%$ less than actual # of total COVID cases on 12/15/2020. QED

2

< see Excel XLSX file upload >

4

$$(P_1, Q_1) = (10, 200)$$

$$(P_2, Q_2) = (15, 150)$$

$$(P_3, Q_3) = (5, Q_3?)$$

a) Slope of Q_3 ?

$$m = \left(\frac{Q_2 - Q_1}{P_2 - P_1} \right) = \left(\frac{150 - 200}{15 - 10} \right) = -10$$

$$y - y_1 = m(x - x_1)$$

$$y - 200 = -10(x - 10)$$

$$y - 200 = -10x + 100$$

$$y = 300 - 10x$$

↓

$$Q_D = 300 - 10P_D$$

$$Q_D(5) = 300 - 10(\$5)$$

$$Q_D(5) = \$250$$

b) $Q_D = 300 - 10P_D$

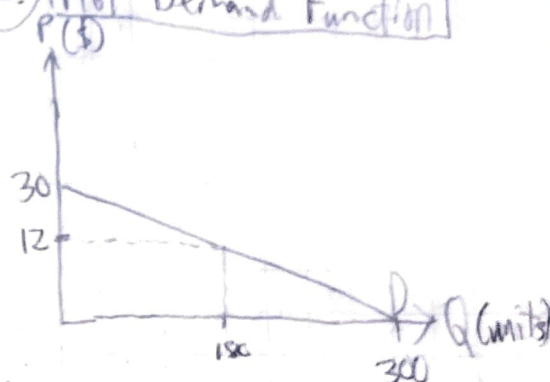
$$10P_D = 300 - Q_D$$

$$P_D = 30 - \left(\frac{1}{10} \right) Q_D$$

$$y\text{-int} = 30$$

$$x\text{-int} = \frac{(-30)}{(-1/10)} = 300$$

c) P_D Demand Function



d) At $Q_D = 180$ units,
will consumers pay
\$10/unit?

$$P_D(180) = 30 - \left(\frac{1}{10} \right) (180)$$

$$P_D(180) = 12$$

According to the Demand Curve, consumers will pay up to \$12/unit for 180 units demanded. Since \$10/unit for 180 units is under the Demand Curve, this is a price consumers will pay for this good. QED

(5) Let

$$\text{income} = M = \$1000$$

$$f = F = q_1$$

$$e = E = q_2$$

$$P_F = P_1 = \$10$$

$$P_E = P_2 = \$25$$

$$(a) M = P_1 q_1 + P_2 q_2$$

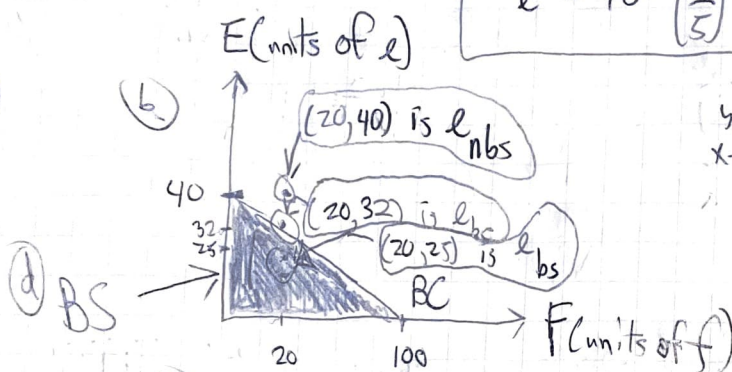
$$\$1000 = (\$10) q_1 + (\$25) q_2$$

$$25 q_2 = 1000 - 10 q_1$$

$$q_2 = \left(\frac{1000}{25} \right) - \left(\frac{10}{25} \right) q_1 = 40 - \frac{2}{5} q_1$$

$$\Downarrow$$

$$e = 40 - \left(\frac{2}{5} \right) f$$



$$y\text{-int} = 40$$

$$x\text{-int} = \frac{-40}{(-2/5)} = 100$$

(c) $\star y\text{-int?}$ If the family spends nothing on food, they can afford to buy 40 units of everything else.

$\star x\text{-int?}$ If the family spends nothing on everything else, they can afford to buy 100 units of food.

$\star \text{slope?}$ For every two units of everything else bought, the family can afford up to five units of food to buy.

(e) Let $f=20$ $e(20) = 40 - \left(\frac{2}{5} \right) (20) = 32$ units of e

	e_{bc}	e_{bs}	e_{nbs}
$f=20$	32	25	40

⑥ a) Time Constraint?

* Measured in units of utility?

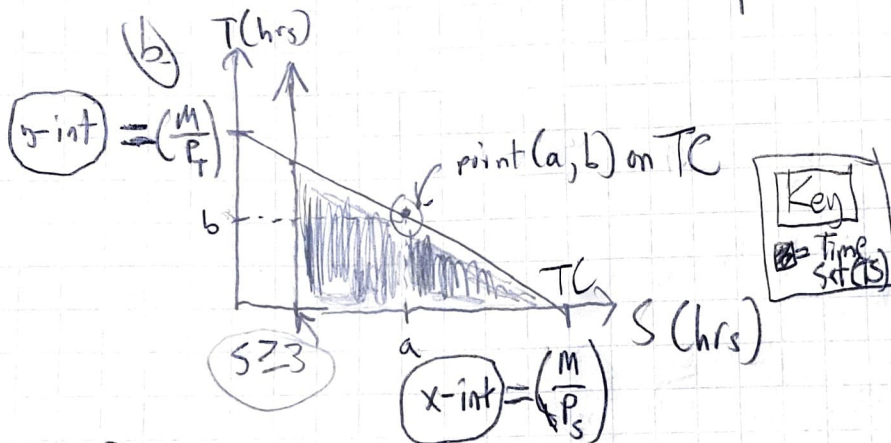
Above a certain value of M ,

- ① Serena will start a fight w/ her Mom →
- ② get grounded & have her phone taken away →
- ③ refuse to do any studying.

So $M = \underbrace{P_S}_{\text{in utils}} S + \underbrace{P_T}_{\text{in utils/hr}} T$ is her Time Constraint for a typical day.

where $P_T \gg P_S$

(we assume she is happier texting versus studying)



(c) slope $= -\frac{P_S}{P_T}$? * Serena experiences strict disutility per hour studying versus strict positive utility per hour texting.

$x\text{-int}_{TC} = \left(\frac{M}{P_S}\right)$? * IF Serena does NO texting it's possible she can spend ALL of her time studying.

$y\text{-int}_{TC} = \left(\frac{M}{P_T}\right)$? * IF Serena does NO studying it's possible she can spend ALL of her time texting.